

Table 1. Performance summary of trajectory inference methods

Methods ranked by composite Friedman rank (rank 1 = best). Mean score across 6 tasks. All metrics: higher = better.

Rank	Method	Marker concordance	Marker monotonicity	kNN smoothness	Root purity	Topology consistency	Composite rank
1	TSCAN	<u>0.334</u>	0.638	0.961	<u>0.726</u>	<u>0.868</u>	2.60
2	Slingshot	0.295	0.618	0.976	0.586	0.825	3.00
3	PAGA + DPT	0.218	0.625	<u>0.991</u>	0.573	0.708	3.20
4	CellRank	0.289	<u>0.753</u>	0.974	0.255	0.553	4.80
5	SCORPIUS	0.284	0.603	0.977	0.160	0.782	4.80
6	Monocle3	0.175	0.575	0.977	0.561	0.611	5.20
7	ElPiGraph	0.203	0.570	0.972	0.512	0.689	6.20
8	VIA	0.166	0.573	0.979	0.486	0.254	6.20

Bold + underline = best value in column. Composite = mean Friedman rank across 5 metrics. kNN smoothness = $1/(1+MAD)$.